

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Regular/Supplementary Winter Examination – 2024

Course: B.Tech.

Branch :Civil Engineering

Semester : V

Subject Code & Name:

Structural Mechanics-II (BTCVC502_Y19)

Max Marks: 60

Date: 08/02/2025

Duration: 3 Hr.

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6.
4. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
5. Use of non-programmable scientific calculators is allowed.
6. Assume suitable data wherever necessary and mention it clearly.

(Level/CO) Marks

Q. 1 Objective type questions. (Compulsory Question)

12

- | | | |
|---|--|---|
| 1 | Static Indeterminacy for pin jointed plane frame is given by | 1 |
| | a. $(m + r) - 2j$ b. $(m + r) - 3j$ c. $(3m + r) - 3j$ d. $(6m + r) - 6j$ | |
| 2 | For drawing ILD, what value of test load is assumed? | 1 |
| | a. Arbitrary b. 1 unit c. Depends upon structure d. 0 | |
| 3 | Calculate the horizontal thrust for the two hinged parabolic arch loaded uniformly throughout with distributed load. | 1 |
| | a. $WL^2/32H$ b. $WL^2/16H$ c. $WL^2/8H$ d. $WL^2/2H$ | |
| 4 | In an Indeterminate structure , when there is a no lack of fit, the partial derivative of strain energy with respect to any of the redundant | 1 |
| | a. will given deflection in the direction of redundant b. will give slope in the direction of redundant c. is a maximum d. is zero | |
| 5 | The shape of the cable, when loaded with uniformly distributed load throughout the span is | 1 |
| | a. Linear always b. Parabolic always c. Trapezoidal always d. Linear or parabolic depending upon the intensity of loading | |
| 6 | $\frac{4EI}{L}$ is stiffness for | 1 |
| | a. far end is hinged b. far end is fixed c. far end is roller support d. None of these | |
| 7 | When a structural member of the uniform section is subjected to a moment at one end only, then the moment required so as to rotate that end to produce a unit slope, is called _____ | 1 |
| | a. Resistance of member b. Stiffness of member c. Capacity of member d. Potential of member | |
| 8 | Stiffness of the end A if the far end B is fixed is _____ | 1 |
| | a. EI/L b. $2EI/L$ c. $3EI/L$ d. $4EI/L$ | |

- 9 Which of the Following methods of structural analysis is a Force Method. 1
 a. Three-moment theorem b. Flexibility matrix method c. Stiffness Matrix method d. None of these
- 10 For beam shown below, the Flexibility Coefficient F_{22} can be written as 1



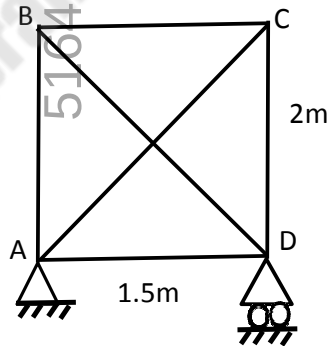
- a. $\frac{L}{3EI}$ b. $\frac{L}{6EI}$ c. $\frac{L}{EI}$ d. None of these
- 11 In one dimensional problem, each node has _____ degrees of freedom 1
 a. 2 degrees of freedom b. 3 degrees of freedom c. no degrees of freedom d. 1 degrees of freedom
- 12 Discretization includes _____ numbering. 1
 a. Element and node b. Only nodal c. Only elemental d. Either nodal or elemental

Q. 2 Solve the following.

12

- A) While fabricating the pin-jointed frame shown in fig. the member AC was the last member to be fitted, and was found to be 1 mm short of the required length. Find the forces in all the members of the frame when the member AC is forced into position. The diagonal members are each 1000 mm^2 in area. While the remaining members are 2000 mm^2 in area. Take $E = 200 \text{ kN/mm}^2$.

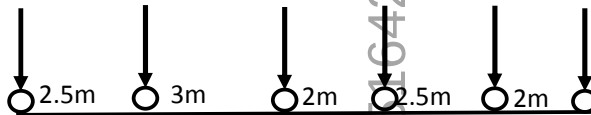
6



- B) The load system shown in fig. moves from right to left on a girder of span 20 m . Find the maximum shear force at a section 7.5m from the left end.

6

$W_1 80\text{kN}$ $W_2 220\text{kN}$ $W_3 100\text{kN}$ $W_4 100\text{kN}$ $W_5 100\text{kN}$ $W_6 100\text{kN}$



Q.3 Solve the following.

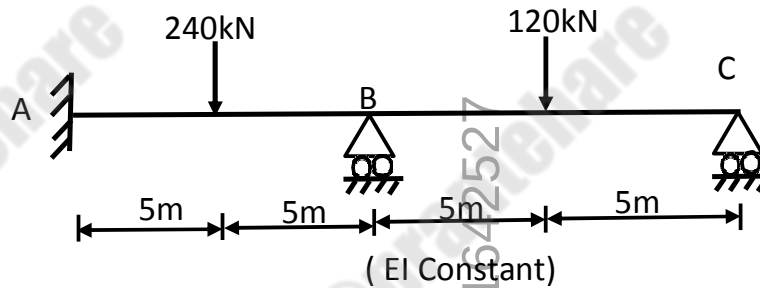
12

- A) A cable is supported between two points 30m horizontally apart. The left support is 3m above the right support. The cable carries a load of 2kN/m on the horizontal span. The lowest point of the cable is 6m below the left support. Find the maximum tension in the cable. 6
- B) A three hinged parabolic arch has span of 20m and a rise of 5m. It carries a UDL of 20 kN per meter over the left half of the span and a point load of 120kN at 5m from the right end. Find the bending moment, normal thrust and radial shear at a section 4m from the left end. 6

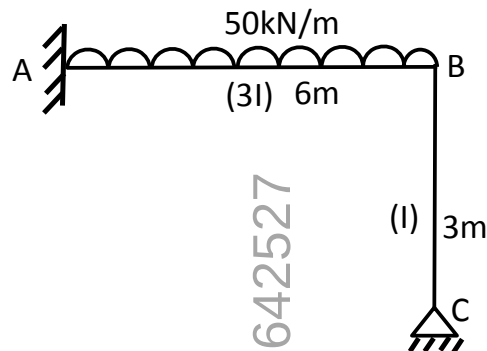
Q.4 Solve Any Two of the following.

12

- A) State and explain the steps for analysis of indeterminate structures as propped cantilever, fixed beam, continuous beam, pin jointed frame and rigid jointed frame by Direct Flexibility method. 6
- B) Analyse the continuous beam shown in Fig. By Force method. 6



- C) Analyse the frame shown in Fig. By Flexibility method. 6



Q.5 Solve Any Two of the following.

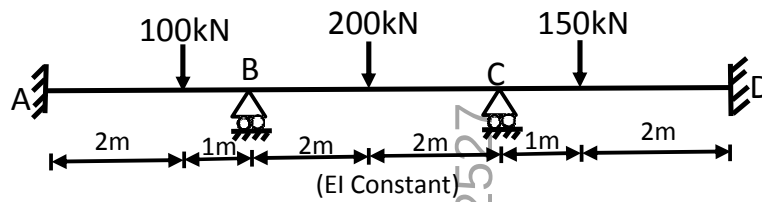
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A) What is the Relation Between the Stiffness matrix and Flexibility Matrix.

6

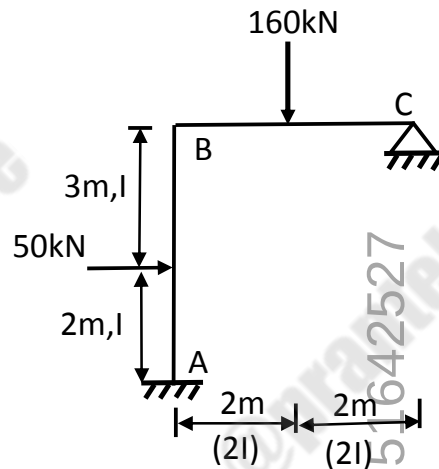
B) Analyse the continuous beam shown in Fig. By stiffness method.

6



C) Analyse the frame shown in Fig. By Displacement method.

6



Q. 6 Solve Any Two of the following.

12

A) Write a note on Pascal's Triangle.

6

B) Explain Plain Stress and Plane Strain problem.

6

C) Explain the term Discretization.

6

*** End ***

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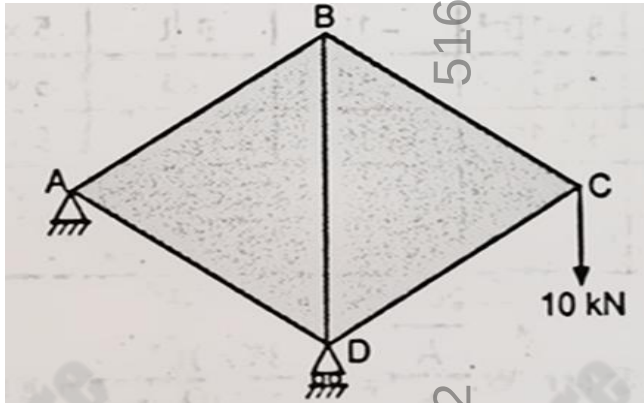
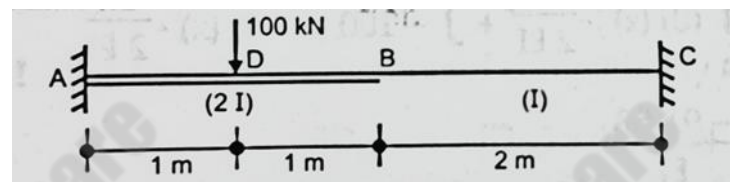
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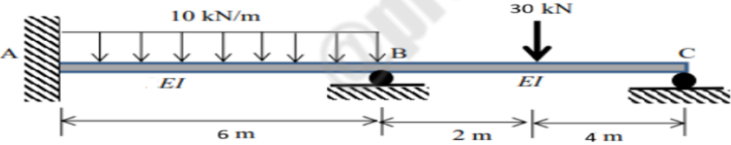
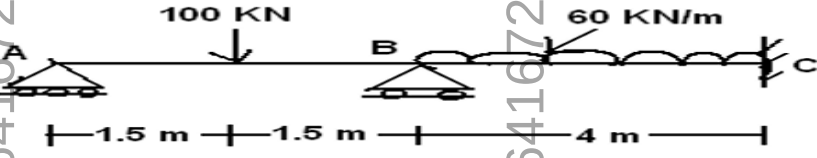
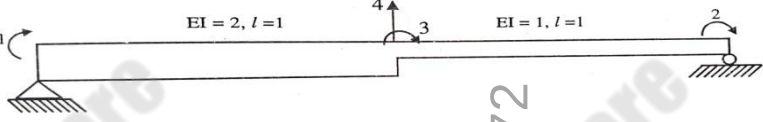
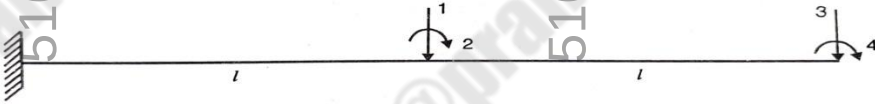
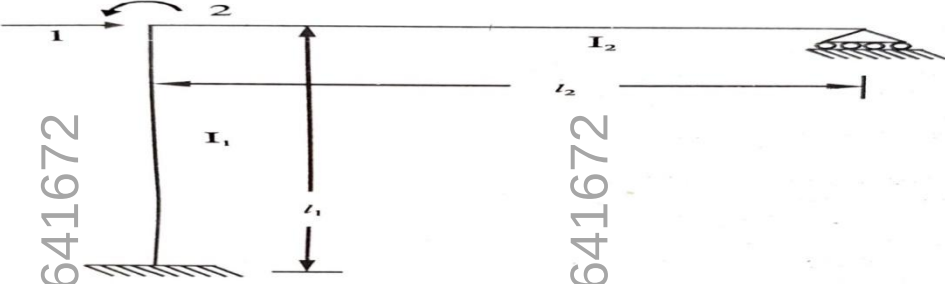
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		(Level/CO)	Marks
Q.1	Objective type questions. (Compulsory Question)		12
1	If there are 'm' unknown member forces, 'r' unknown reaction components and 'j' number of joints, then degree of static indeterminacy of a pin jointed plane frame is given by a. $m + r + 2j$ b. $m - r + 2j$ c. $m + r - 2j$ d. $m + r - 3j$	1	1
2	If there are 'm' unknown member forces, 'r' unknown reaction components and 'j' number of joints, then a pin jointed plane frame is unstable if a. $(m + r) > 2j$ b. $(m + r) = 2j$ c. $(m + r) < 2j$ d. None of the above	1	1
3	For drawing ILD, what value of test load is assumed? a. 1 Unit b. Arbitrary c. Depends upon structure d. 0	1	1
4	ILD of statically determinate beams consists of curve as well as straight lines. a. True b. False	1	1
5	Cable is a tension member a. True b. False	2	1
6	The shape of the cable, when loaded with uniformly distributed load throughout the span is a. Linear always b. Parabolic always c. Trapezoidal always d. Linear or parabolic	2	1
7	Which of the following conditions form the basis of Direct Flexibility Method a. Equilibrium Conditions b. Force-Displacement conditions c. Load Applied d. Compatibility Conditions	3	1
8	Flexibility coefficients are used in which of the following method? a. Force method b. Displacement method c. Both force and displacement method d. Virtual force method	3	1
9	Stiffness coefficients are used in which of the following method? a. Force method b. Displacement method c. Both force and displacement method d. Virtual force method	4	1
10	Which of the following conditions form the basis of Direct Stiffness Method a. Equilibrium b. Force- c. Load Applied d. Compatibility	4	1

	Conditions	Displacement conditions		Conditions		
11	If the structure is divided into discrete areas or volumes then it is called as				5	1
	a. Structure	b. Element	c. Matrix	d. Boundaries		
12	One dimensional element is the linear segments which are used to model				5	1
	a. Bars and trusses	b. Plates and beams	c. Structures	d. Solids		
Q. 2	Solve the following.					12
A)	A truss consisting of two equivalent triangles is loaded as shown in figure. Determine the horizontal deflection of roller at D. The length of each member is 3m. The C.S. areas of all ties are 300mm ² each and that of all struts are 600mm ² each. Take E= 200GPa.				1	6
						
B)	A simply supported beam CD has span 12m. It is loaded with UDL 45kN/m, span 5m, and starts 3.5m from C. A concentrated load of 50kN is acting at 2m from D. Find Shear Force at 5m from C by ILD method.					6
Q.3	Solve the following.					12
A)	A light cable is supported at two points 20m apart, which are at same level. The cable supports three equidistance point loads of 40kN, 30kN and 20kN from left support. The deflection at first point is found to be 0.8m. Determine the tension in the different segments and total length of Cable.				2	6
B)	A three hinged parabolic arch of 20m span and 3m rise is carrying a point load of 100kN at a section 5m from the left support. Find the value of horizontal thrust and bending moment at a point 7.5m from the right support.				2	6
Q. 4	Solve Any Two of the following.					12
A)	Analyze the beam shown in figure by flexibility method.				3	6
						

B)	Analyze the beam shown in figure by flexibility method if support B sinks by 25mm. Take $EI=3800\text{KNm}^2$	3	6
			
C)	Analyse the beam shown in figure by flexibility method.	3	6
			
Q.5	Solve Any Two of the following.		12
A)	Develop the stiffness matrix of the beam shown with respect to the 04 degrees of freedom given.	4	6
			
B)	Develop the stiffness matrix of the beam shown with respect to the 04 degrees of freedom given.	4	6
			
C)	Develop the stiffness matrix of the frame shown with respect to the given coordinates.	4	6
			
Q. 6	Solve Any Two of the following.		12
A)	Write short note on types elements used in finite elements analysis method.	5	6
B)	Write short note on Plane Strain and Axi- symmetric Problem in finite elements analysis method.	5	6
C)	Write short note on shape functions used in finite elements analysis method.	5	6
*** End ***			